

MORPHOLOGY OF PUERULUS AND EARLY JUVENILE STAGES OF THE SPINY LOBSTER *PALINURUS MAURITANICUS* (DECAPODA: PALINURIDAE)

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ABSTRACT

Puerulus and early juvenile stages of the spiny lobster *Palinurus mauritanicus* Gruvel, 1911 from the western Mediterranean are described and illustrated, and their morphological characteristics are compared with those described in other palinurids. This constitutes the first detailed morphological description of a puerulus of the genus *Palinurus*. The puerulus of *P. mauritanicus* presents a carapace already with some spines characteristic of the adult phase, which allows to differentiate it from the congeneric species *Palinurus elephas* (Fabricius, 1787) co-occurring in most of its distribution area. The puerulus of *P. mauritanicus* can be differentiated from those described for pueruli of the genera *Panulirus* and *Jasus* by the presence of very long, flagellated, and setose exopods on maxillipeds 2 and 3. The puerulus of *P. mauritanicus* resembles those described for *Projasus* by the presence of the crista dentata on the third maxilliped. The morphology and differential colouration of the early juvenile with respect to the adult are described.

INTRODUCTION

The last phyllosoma instar of palinurid lobsters metamorphoses into a further planktonic stage, the puerulus (Gurney, 1942; Booth and Phillips, 1994), which is the stage in charge of performing the settlement or transition from the planktonic to a benthic existence (Phillips and Olsen, 1975; Herrnkind et al., 1994; Jeffs and Holland, 2000; Lipcius and Eggleston, 2000). After settlement, the puerulus metamorphoses into the adult-like juvenile (Crosnier, 1971; Cooke and McDonald, 1981; Báez and Ruiz, 2000).

The puerulus and early juvenile morphology of the five species of the genus *Palinurus* is only partially known for *Palinurus elephas* (Fabricius, 1787) (Bouvier, 1913, 1914; Santucci, 1926; Orton and Ford, 1933; Caroli, 1946; Gordon, 1953; Hunter, 1999). However, more information is available on the pueruli of the genera *Panulirus*, *Jasus* and *Projasus* (Gordon, 1953; Berry, 1974; Deshmukh, 1966; Nishida et al., 1990; Briones-Fourzán and McWilliam, 1997; Kittaka et al., 1997; Báez and Ruiz, 2000).

Three species of palinurid spiny lobster are distributed in the northeastern Atlantic and Mediterranean Sea (Zariquiey-Álvarez, 1968; d'Udekem d'Acoz, 1999): *Palinurus elephas* (Fabricius, 1787), *Palinurus mauritanicus* Gruvel, 1911, and *Panulirus regius* de Brito Capello, 1864, which is recorded in coastal Atlantic areas off Northwest Africa, with scattered occasional records in the Mediterranean Sea (Gibert i Olivé, 1920; Zariquiey-Cenarro, 1935; Galil et al., 2002). *P. mauritanicus* occurs along the western coasts of Africa and Europe from Senegal to the western coasts of Ireland, and throughout the Mediterranean Sea (Zariquiey-Álvarez, 1968; Minchin, 1989; d'Udekem d'Acoz, 1999), but not in the Adriatic (Stevic, 1990; Holthuis, 1991). Bathymetrically, this species has been recorded from around 40 to 750 m but preferentially between around 150 and 400 m (Maurin, 1968; Relini et al., 1986; Abelló et al., 1988,

2002; Ceccaldi and Latrouite, 2002). It has been recorded as a characteristic species of deep-water coral reefs and muddy substrates near rocky outcrops (Minchin, 1989; Secchi and Cau, 1999).

The present study aims to describe the morphology of the puerulus and early juvenile stages of *P. mauritanicus* and to compare their morphological features with those known for other species of palinurid spiny lobsters.

MATERIALS AND METHODS

Two pueruli of *Palinurus mauritanicus* were obtained by demersal trawling on muddy bottoms of the continental slope off Catalonia in the western Mediterranean (Specimen A: 41°54.7'N, 3°30.8'E; 409–452 m depth; 29 May 2004; trawl number MEDITS_ES_04 L107. Specimen B: 40°56.4'N, 1°25.7'E; 329–339 m depth; 27 May 2005; trawl number MEDITS_ES_05L080). Two early juveniles of *Palinurus mauritanicus* were also captured by demersal trawling on muddy bottoms of the continental slope off the Ebro river delta in the western Mediterranean (Specimen C: 25 June 2000, 40°56'42"N, 1°31'03"E, 420–427 m, trawl number NERIT2 N2T23; Specimen D: 23 June 2000, 40°56'30"N, 1°31'34"E, 453–459 m, trawl number NERIT2 N2T05).

All individuals were preserved in 70% ethanol. A binocular microscope equipped with an ocular micrometer was used for the dissection and measurements of individuals. A microscope was used for the determination of the setation and measurements of some appendages. All drawings were made with the aid of a camera lucida and microscope photography. The following measurements were taken: carapace length (CL) from the rostral margin to the posterior margin of the carapace; total length (TL) from the rostral margin to the posterior margin of the telson; length of antennule peduncle segments (AP1, AP2, AP3); outer antennular flagellum length (OFL); inner antennular flagellum length (IFL); pleopod length (PL) from the basal margin of the protopod to the distal margin of the exopod. Since the antennal flagella of the studied individuals was broken, probably due to the trawling process, total antennal length has not been measured. Since carapace lateral spinulation is symmetrical, it is described only from one side of the animal. The stages are described using the basic malacostracan somite plan from anterior to posterior and appendage segments are described from proximal to distal, endopod then exopod (Clark et al., 1998). When the type of seta is stated, terminology has followed Garm (2004).

Unless otherwise stated, all measurements and meristic counts of the early juvenile description are based on specimen C.

All four specimens studied have been deposited in the Biological Collections of Reference of the Institut de Ciències del Mar (CSIC) in Barcelona (Codes: ICMD_9/2005, ICMD_1/2006, ICMD_10/2005 and ICMD_11/2005, respectively for specimens A, B, C and D).

RESULTS

The puerulus stage is described in detail, while in early juvenile stages only differences and changes are described.

Puerulus

Size.—TL = 23.3–24.0 mm; CL = 8.8–9.0 mm.

Colour.—Transparent, except brownish eyes.

Carapace (Fig. 1A, B).—Rostrum present and rudimentary. One large supraorbital spine plus two dorsal spines immediately behind. Three anterolateral spines: a large suborbital spine, plus two smaller spines immediately behind. Anterior branchial region with 3–4 spines. Gastric region with 2 longitudinal rows of 3 spines. Cervical groove slightly marked. Cardiac region with 2 rows of 3 post-cervical spines. A row of 5 dorsolateral spines present. Length of the antennular plate 5.5% of CL.

Sternum (Fig. 1B).—Cuticle thin and transparent and clearly shows the thoracic ganglia through it. The posterolateral angles of fused sternites 2–5 bear one straight spine each, ventrally directed, the smallest being in the second sternite and increasing in size posteriorly.

Antennule (Figs. 1A, 2A).—Well developed; projecting slightly beyond distal segment of antennal peduncle. Peduncle is longer than inner and outer flagellum. Inner flagellum 16-segmented, longer than the outer, with dense rows of setae; outer flagellum 13-segmented, more robust than the inner, with dense rows of setae and aesthetascs. AP1 = 2.25 mm, AP2 = 0.68 mm, AP3 = 1.63 mm, OFL = 1.7 mm, IFL = 2.5 mm.

Antenna (Fig. 1A).—Peduncular spines as illustrated. Plectrum present on basal segment of the peduncle.

Mandible (Fig. 2B).—Incisor and molar processes undifferentiated, uncalcified and lacked of any teeth. Palp unsegmented and without setae.

Maxillule (Fig. 2C).—Poorly developed, with very reduced setation. Coxal endite with 2–3 terminal minute spines. Basal endite with 4–5 terminal minute spines and 6 minute setae. Endopod unsegmented, with 1 subterminal simple seta.

Maxilla (Fig. 2D).—Three endites present with 5–8, 5–6, 9–11 sparsely setose setae; the distal endite elongated, longer than the others. Endopod unsegmented with 6–9 pappose setae on outer margin. Scaphognathite (exopod) large and well developed, with about 145 plumose marginal setae.

First Maxilliped (Fig. 3A).—Epipod well developed, with about 16 long simple setae. The unsegmented endopod and endites rudimentary, without setae. Exopod large and flagellated, with about 15 proximal plumose setae on outer margin; flagellum with about 22 segments, more differentiated distally; each segment with 2 setae (minute proximally; long and plumose distally).

Second Maxilliped (Fig. 3B–B₁).—Gill present and well developed. Epipod well developed, with about 20 long simple setae. Endopod 4-segmented, without setae except on propodus, with one minute distal seta, and dactylus, with 5 minute spines and 3 minute setae. Exopod long and flagellated, approximately twice the endopod length; flagellum with 21–22 segments more differentiated distally; each segment with two setae (minute proximally; long and plumose distally).

Third Maxilliped (Fig. 3C, D).—Gill present and well developed. Epipod well developed, with about 30 long simple setae. Coxa with 14–16 simple setae and one tooth on inner margin. Basis with 3 minute setae. Endopod 5-segmented. Ischium with sparse small setae (about 18 simple setae) and 4–7 teeth on inner margin (crista dentata); merus with sparse small setae (about 14–15 simple setae) and 2 distal serrate setae; carpus with sparsely minute setae and about 35 serrate setae; propodus with sparse minute setae and about 16 serrate setae; dactylus with 10–15 setae (4 serrated, 5–10 cuspidate and one simple setae). Exopod large and flagellated, reaching beyond base of propodus; flagellum with 15 segments more differentiated distally, each segment with two setae (minute proximally; long and plumose distally).

Pereiopods (Figs. 1B, 4A–E).—Pereiopods 1–5 with segmented and well developed endopod; pereiopods 1–4 with reduced and vestigial exopods.; pereiopods 1–4 with one coxal spine, pereiopod 5 with 3 coxal spines; merus of pereiopods 1–5 with a distal marginal spine dorsally. Length of the segments of pereiopods 1–5 (merus to dactylus) as shown in Table 1. Pereiopod 1 is the shortest; third and fourth pereiopods are the longest.

Pleon (Fig. 1A, B).—Somites with smooth tergites; transverse groove present (slightly marked) in somites 1–4 (see Fig. 1B); pleural plates produced into a strong spine directed ventrally, posterior edge of plates of somites 2–6 with one simple spine. Pleura of somite 2 with a poorly marked anterior tooth. Somite 6 with one pair of dorsal spines.

Pleopods (Fig. 5).—Somites 2–5 each with one pair of well developed biramous pleopods. Endopod unsegmented with about 55–61 long marginal plumose setae and appendix interna with 2–3 distal setae on outer margin and 11–12 terminal coupling hooks on inner margin. Exopod

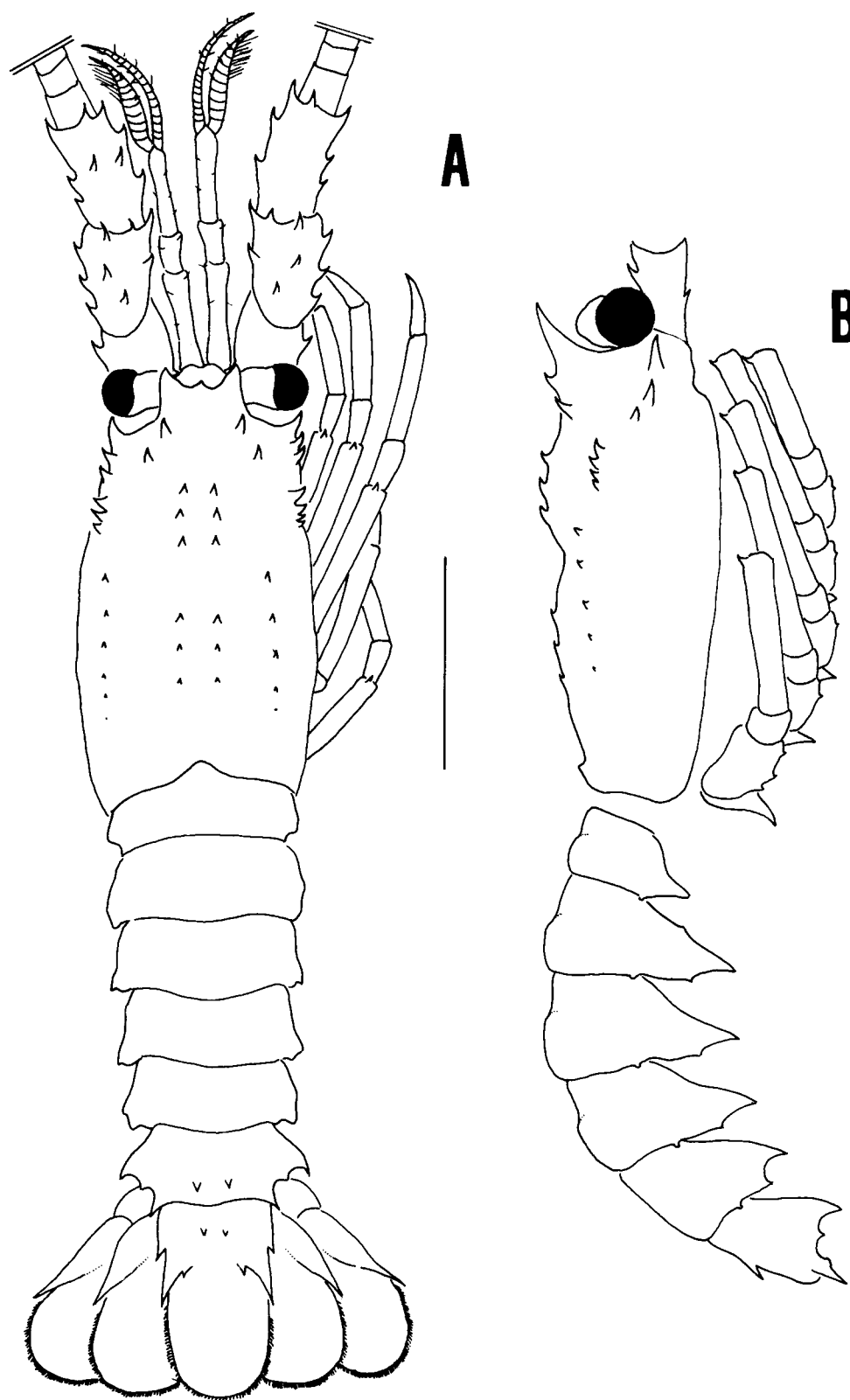


Fig. 1. *Palinurus mauritanicus* Gruvel, 1911. Puerulus. A, dorsal view; B, lateral view. Scale bars = 5 mm.

unsegmented with about 46-49 long marginal plumose setae. PL4 = 3.3 mm.

Uropods (Fig. 1A).—Biramous; with one spine on outer margin.

Telson (Fig. 1A).—With one pair of dorsal spines; the lateral margin of the calcified part bears one pair of spines; the posterior margin bears 2 pairs of spines; the posterior margin of the uncalcified part setose.

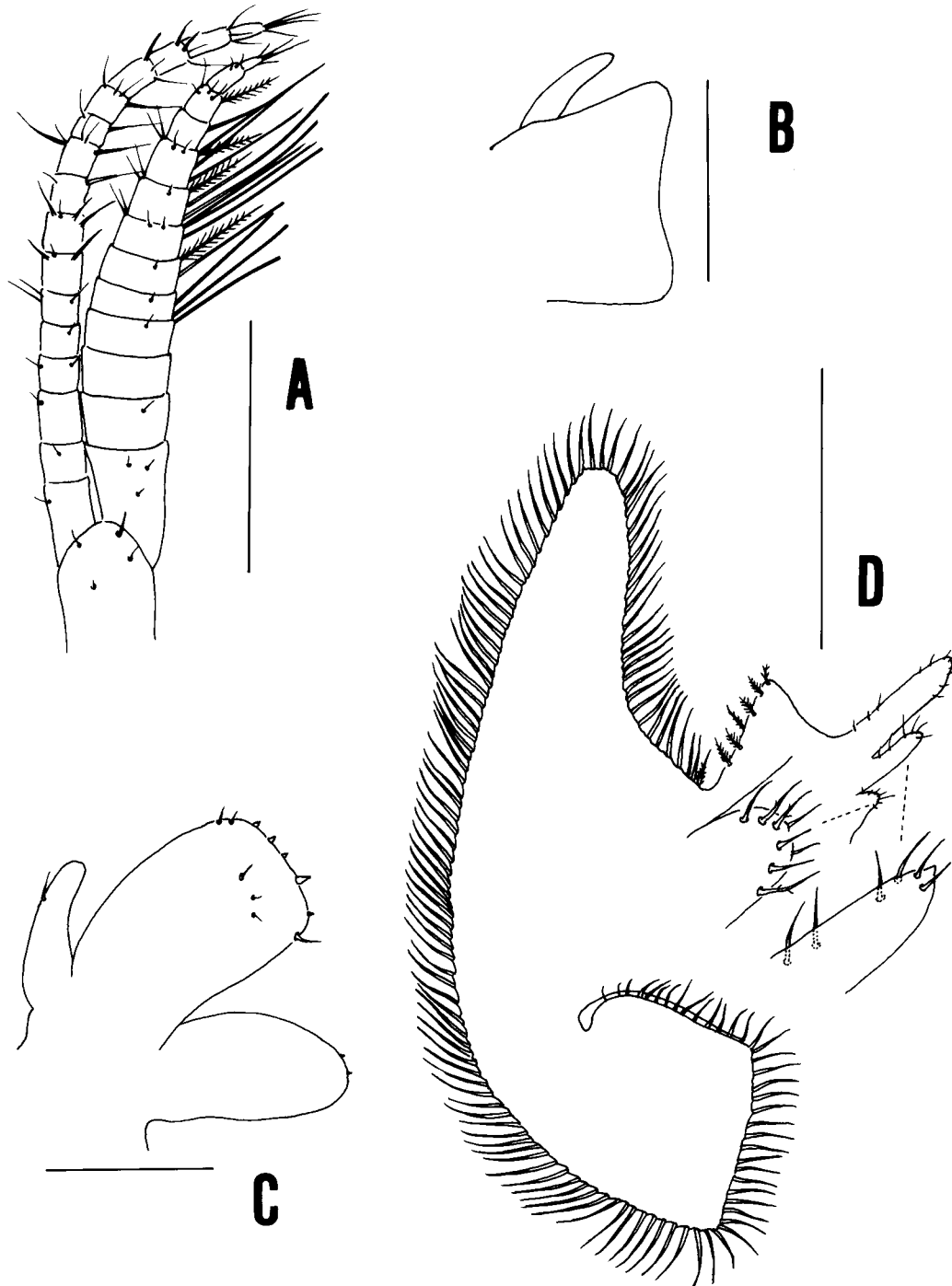


Fig. 2. *Palinurus mauritanicus* Gruvel, 1911. Puerulus. A, antennal flagellum; B, mandible; C, maxillule; D, maxilla; setulae of the plumose setae of scaphognathite have been omitted. Scale bars of A, B and D = 1 mm, scale bar of C = 0.5 mm.

Early Juvenile

Size.—TL = 24.1 mm (specimen C), 28.0 mm (specimen D); CL = 9.2 mm (specimen C), 10.2 mm (specimen D).

Colour.—A dorsal brownish-reddish dark band extends dorsally from the frontal and gastric region along the carapace and pleon, excluding the telson. This dark band is

flanked laterally by a light band on each side of the dorsal carapace and pleon.

Carapace (Fig. 6A, B).—Spinulation and setation more developed than in puerulus and similar to the adult phase. Cervical groove more marked than in puerulus, similar to adult. Internal margin of supraorbital spines spinulated. Space between supraorbital spines not yet as

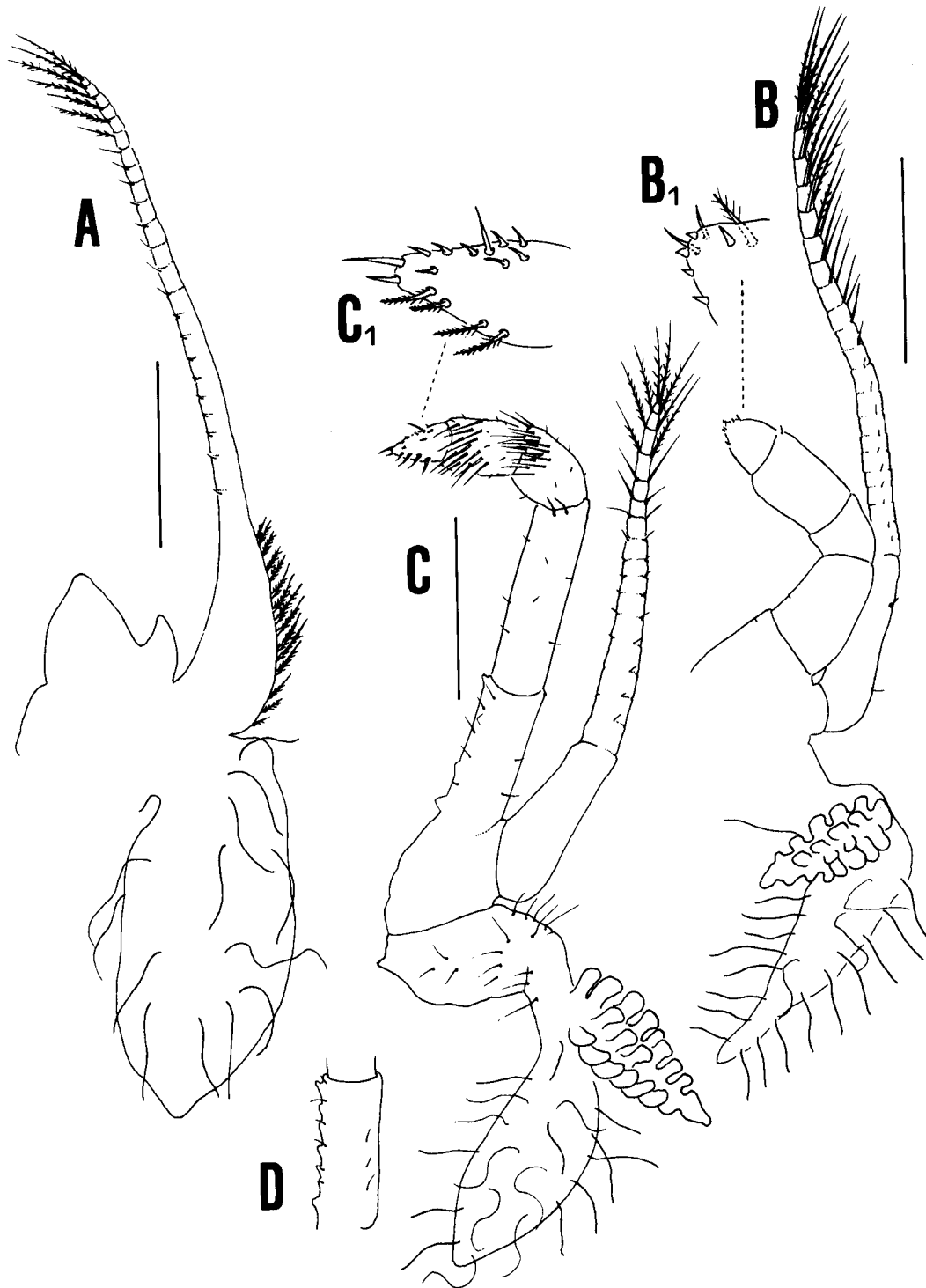


Fig. 3. *Palinurus mauritanicus* Gruvel, 1911. Puerulus. A, maxilliped 1; B, maxilliped 2; B₁, maxilliped 2, detail of distal part of the dactylus; C, maxilliped 3; C₁, maxilliped 3, detail of distal part of the dactylus; D, maxilliped 3, detail of the ischium of the puerulus B. Scale bars = 1 mm.

broad as in adult. Antennular plate longer than puerulus stage: 12.7 % of CL.

Sternum (Figs. 6B, 10B).—Posterolateral spines shorter and not as acute as in puerulus. Pairs of spines appear as shown in Fig. 10B.

Antennule (Fig. 6A).—Similar to puerulus (see Fig. 2A). Inner flagellum with 18 segments, and outer flagellum with 15 segments. AP1 = 3.12 mm, AP2 = 0.92 mm, AP3 = 1.60 mm, OFL = 2.25 mm, IFL = 2.8 mm (measurements for specimen C).

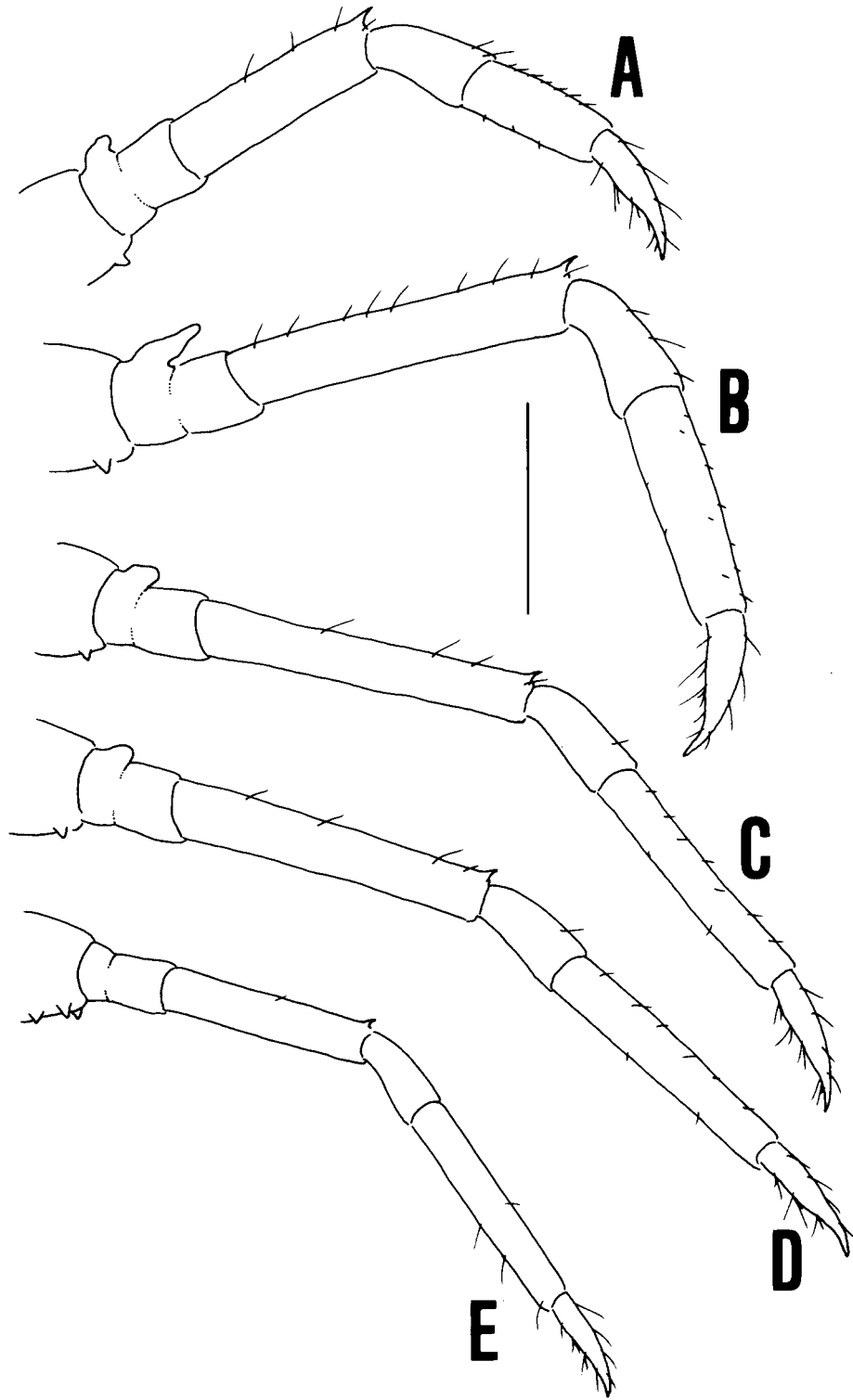


Fig. 4. *Palinurus mauritanicus* Gruvel, 1911. Puerulus. Pereiopods. A-E, pereopods 1-5. Scale bar = 3 mm.

Antenna (Fig. 6A).—Peduncular spines as illustrated. Plectrum more developed than in puerulus stage.

Mandible (Fig. 7A, A₁).—Well developed and calcified. Palp 3-segmented with 0,7,8 setae; second segment with distal surface spinulated.

Maxillule (Fig. 7B).—Setation much more developed than in puerulus. Coxal endite with 32 setae. Basal endite with 29 setae. Endopod unsegmented, with 12 setae.

Maxilla (Fig. 7C).—Similar to puerulus; endopod with 20 pappose setae; endites elongated with 6,11,24 setae.

Table 1. Length (in mm) of the pereopods of puerulus (specimen A) and early juvenile (specimen C) of *Palinurus mauritanicus* Gruvel, 1911.

	Pereopod				
	1	2	3	4	5
Puerulus					
merus	3.00	4.0	4.25	4.0	2.90
carpus	1.15	1.5	1.50	1.5	1.25
propodus	1.50	2.5	2.75	3.0	2.75
dactylus	1.60	1.4	1.60	1.7	1.60
total	7.25	9.4	10.10	10.2	8.50
Juvenile					
merus	3.50	4.00	4.25	4.50	3.50
carpus	1.25	1.25	1.30	1.25	1.25
propodus	2.00	2.50	3.25	3.50	2.70
dactylus	1.60	1.50	1.65	1.95	1.60
total	8.35	9.25	10.45	11.20	9.05

First Maxilliped (Fig. 8A).—Endopod unsegmented with 19 setae. Endite with about 40 setae. Exopod with about 37 proximal setae.

Second Maxilliped (Fig. 8B).—Setation of the endopod well developed (more setose than in puerulus), ischium with 22 setae, merus with 9 minute setae, propodus with about 38 setae, and dactylus with 13 cuspidate setae plus 3 serrulate setae.

Third Maxilliped (Fig. 9A).—Surface very spinulated. Basis with 3 acute processes and about 20 setae in the inner margin. Endopod with two rows of stout teeth on the inner margin of the ischium (7 and 8 teeth) and merus (6 and 11 teeth) (crista dentata). Setation of the endopod more developed than in puerulus; ischium with about 25 setae on inner margin and 11 setae on outer margin; merus with about 29 setae on inner margin and 16 setae on outer margin; carpus with more than 50 serrate setae; propodus with more than 50 serrate setae; dactylus with 15 cuspidate and 5 serrate setae.

Pereopods.—Without exopods; similar in shape to those in puerulus, but slightly longer (Table 1), more setose and more robust, especially pereopod 1; (see Fig. 10A). The first pereopod is the shortest followed by pereopods 2 and 5, of similar length; pereopod 3 is long but not as long as pereopod 4, which is the longest.

Pleon (Fig. 6C).—Posterior margins of somites setose. Pleonal tergites of somites 1–3 with marked grooves, slightly grooved in somite 4. Pleura of somite 2 with a strong anterior tooth. Posterior margin of pleura of somites 2–4 with 1–2 additional small teeth.

Pleopods (Fig. 9B).—Very much reduced than in puerulus (PL4 = 1.4 mm). Endopod smaller than exopod, with only 3 proximal setae; appendix interna without terminal hooks. Exopod with about 40 long marginal plumose setae, 8 distal subterminal cuspidate setae and 2 cuspidate setae on proximal surface.

Uropods (Fig. 6D).—With additional spines on posterior margin of the calcified part.

Telson (Fig. 6D).—Additional spinulation and setation appears on the dorsal surface.

DISCUSSION

The puerulus is the post-larval transitional stage between the planktonic phyllosoma and the benthic juvenile, and resembles in morphology the adult phase (Gurney, 1942; Booth and Phillips, 1994; McWilliam, 1995). However, the carapace of the puerulus stage lacks the grooves, setation and most of the spines of the adult, and the appendages present important morphological differences (Gordon, 1953; Michel, 1971; Crosnier, 1971; Cooke and McDonald, 1981; Báez and Ruiz, 2000). In many pueruli descriptions, some characters of ontogenetic significance are frequently poorly figured or not specifically mentioned in the text, which difficulties comparison between species.

The key characteristics, useful for diagnosis, of the puerulus of *Palinurus mauritanicus* concern spination of the carapace such as the three pairs of spines on the gastric region and the three anterolateral spines; the presence of a rudimentary rostrum; the antennular flagella shorter than the peduncle, the outer being more robust than the inner; the second segment of the antennular peduncle shorter than the third; the exopod of maxilliped 2 and 3 long, with a multi-segmented flagellum that bears plumose setae.

Other characters are exclusive of this developmental stage associated to the behavioural role between pelagic and benthic life, and are present in all described pueruli, such as the presence of well-developed pleopods with appendices internae bearing coupling hooks (Báez, 1983; Jeffs and Holland, 2000; Inoue et al., 2002), the incomplete development of the mouthparts, with the mandible lacking calcification and the palp lacking setae, and the maxillule, maxilla and maxillipeds being similar in segmentation to the early juvenile, but much less setose (Briones-Fourzán and McWilliam, 1997; Lyons and Hunt, 1997; Inoue et al., 2002).

The accurate comparison of the puerulus of *P. mauritanicus* with that of *P. elephas* cannot be perfectly achieved since previous descriptions of *P. elephas* are not complete and do not follow the modern standards; the mouth parts are not described in detail or are not described on the text. The puerulus of *P. mauritanicus* present similar morphological features to those described for *P. elephas*: the antennule projecting slightly beyond the distal segment of the antennal peduncle; the antennular flagella shorter than the peduncle, the outer more robust than the inner; the second segment of the antennular peduncle shorter than the third; presence of rostrum; presence of three anterolateral spines, and two pairs of dorsal spines behind the supraorbital spines (Gordon, 1953). Santucci (1926) showed a drawing of the anteroventral region of the carapace of the puerulus of *P. elephas* with clearly long, and flagellated maxilliped exopods similar to those described presently for *P. mauritanicus*.

The main differences between the pueruli of *P. mauritanicus* and *P. elephas* are size and spinulation of the carapace. Total length is larger in *P. mauritanicus* than

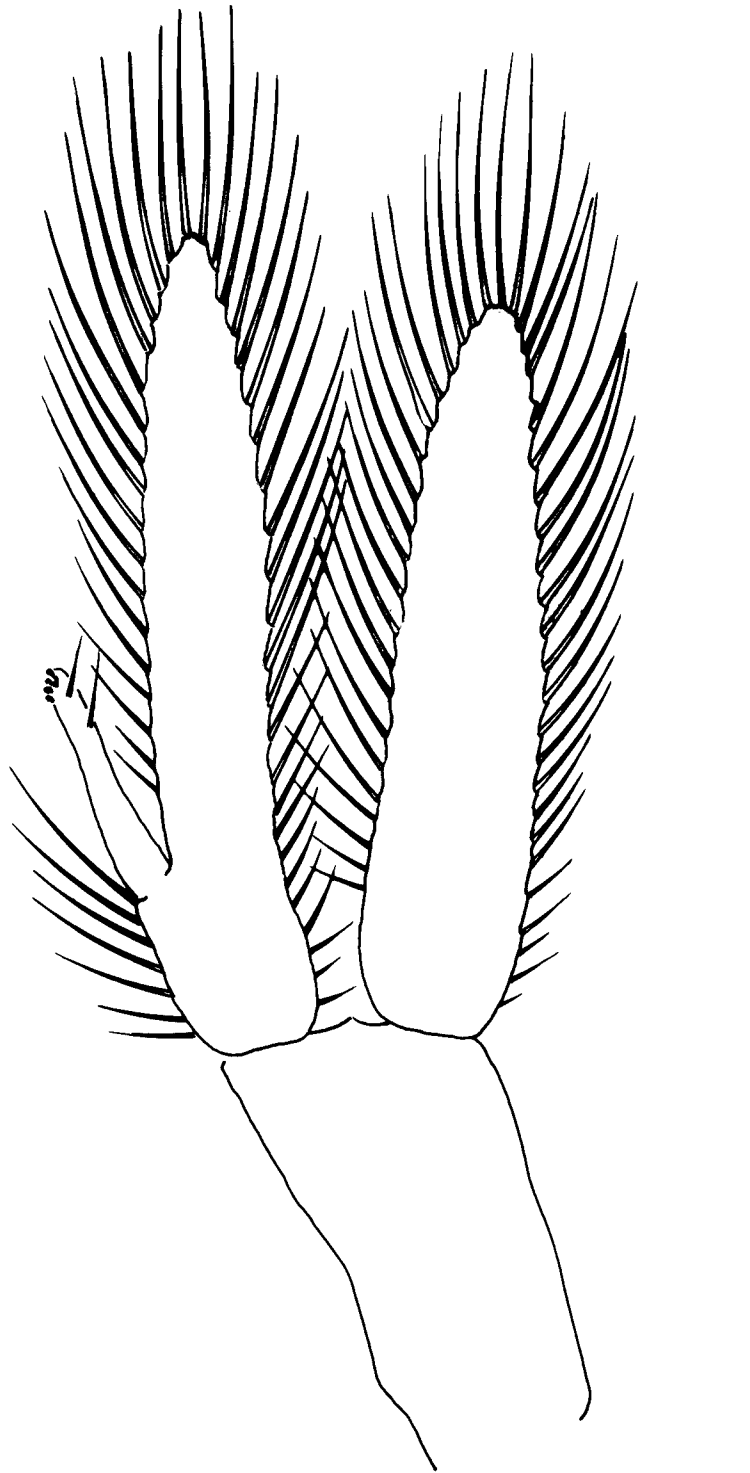


Fig. 5. *Palinurus mauritanicus* Gruvel, 1911. Puerulus. Pleopod 4. Setulae of the plumose setae have been omitted. Scale bar = 1 mm.

in *P. elephas* (24 mm TL versus 21 mm TL; Hunter, 1999). *P. mauritanicus* presents two rows of 3 spines in the gastric region (the arrangement and size of the spines is different from that shown by *P. elephas*; Gordon, 1953). The published descriptions and drawings of puerulus of *P. elephas* (Bouvier, 1914; Santucci, 1926; Fage, 1927) do not show the occurrence of the two dorsal spines on abdominal somite

6 present in *P. mauritanicus*. However, this characteristic may have been overlooked since they are present in the early juvenile (Orton and Ford, 1933). In other descriptions of palinurid pueruli, these spines have only been described in the puerulus of *Projasus bahamondei* and *P. parkeri* (Webber and Booth, 1988; Báez and Ruiz, 2000), and are absent from *Panulirus*.

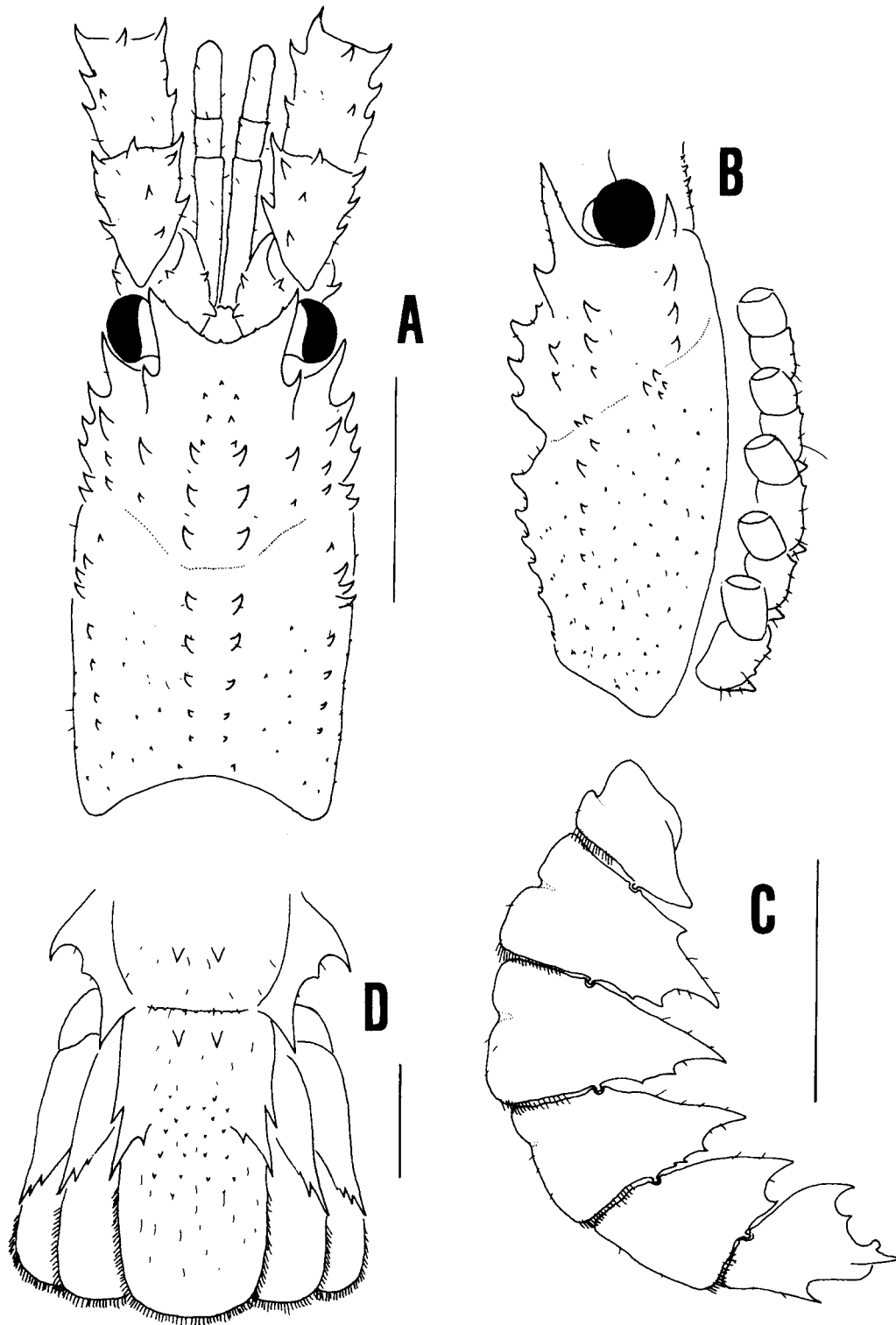


Fig. 6. *Palinurus mauritanicus* Gruvel, 1911. Early juvenile. A, cephalothorax, dorsal view; B, cephalothorax, lateral view; C, pleon, lateral view; D, telson and uropods, dorsal view. Scale bars = 5 mm.

As in adults, the puerulus of *P. mauritanicus* can be differentiated from those of *Panulirus* species by the length of the antennular flagella (shorter than the peduncle in *P. mauritanicus*) (Goldman and Patek, 2002). Another important feature is the presence of the long flagellated

exopod in maxillipeds 2 and 3 in *P. mauritanicus*. The exopod of the second maxilliped in *Panulirus* reaches beyond the propodus of the endopod, whereas it clearly surpasses the dactylus in both *P. mauritanicus* and *P. elephas*. The exopod of the third maxilliped of *Panulirus*

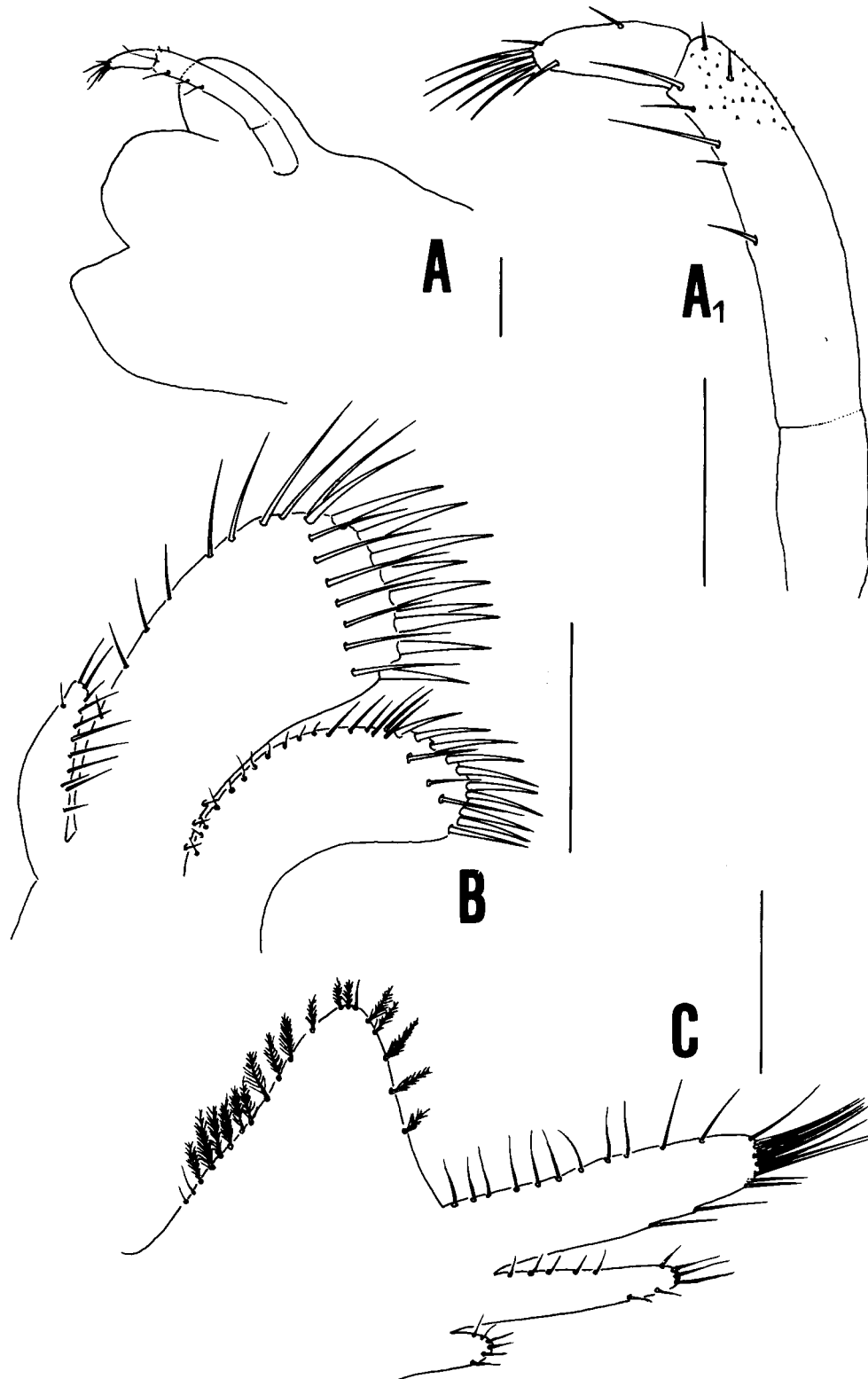


Fig. 7. *Palinurus mauritanicus* Gruvel, 1911. Early juvenile. A, mandible; A₁, mandibular palp; B, maxillule; C, maxilla (without the scaphognathite). Scale bars = 0.5 mm.

is vestigial or reaches or goes slightly beyond the merus of the endopod (McWilliam, 1995), whereas it reaches or goes slightly beyond the propodus in *P. mauritanicus*. Also, *P. mauritanicus* shows a row of teeth on the ischium

of the third maxilliped which represent a simple crista dentata, a character already developed similarly to the adult in the early juvenile stage. A row of teeth on the ischium of the third maxilliped appears also on the figures

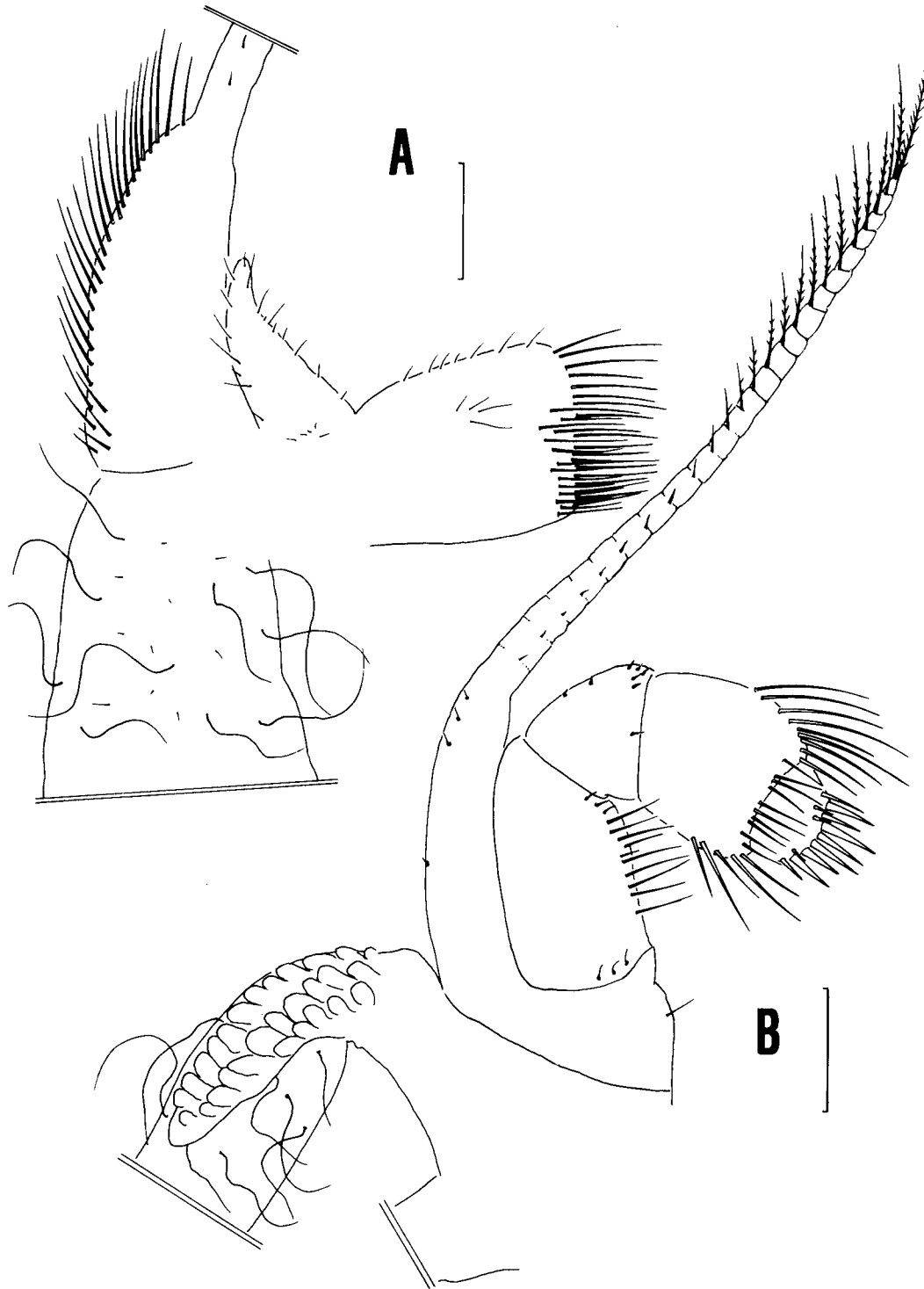


Fig. 8. *Palinurus mauritanicus* Gruvel, 1911. Early juvenile. A, first maxilliped; B, second maxilliped. Scale bars = 0.5 mm.

presented by Santucci (1926) for the puerulus of *P. elephas*. The crista dentata are also present in the puerulus stage of *Projasus bahamondei* and *P. parkeri* (Webber and Booth, 1988; Báez and Ruiz, 2000) (Table 2).

The genus *Panulirus* is considered the most recent within the Palinuridae and contains the greatest number of species (McWilliam, 1995). Within this genus the

reduction of the second and third maxilliped exopods is considered a derived character within the genus (George and Main, 1967; McWilliam, 1995; Patek and Oakley, 2003). George and Main (1967) established four adult groups of *Panulirus* species based on the comparative morphology of these appendages. Additionally, the puerulus morphology corresponds well with the *Panulirus* adult

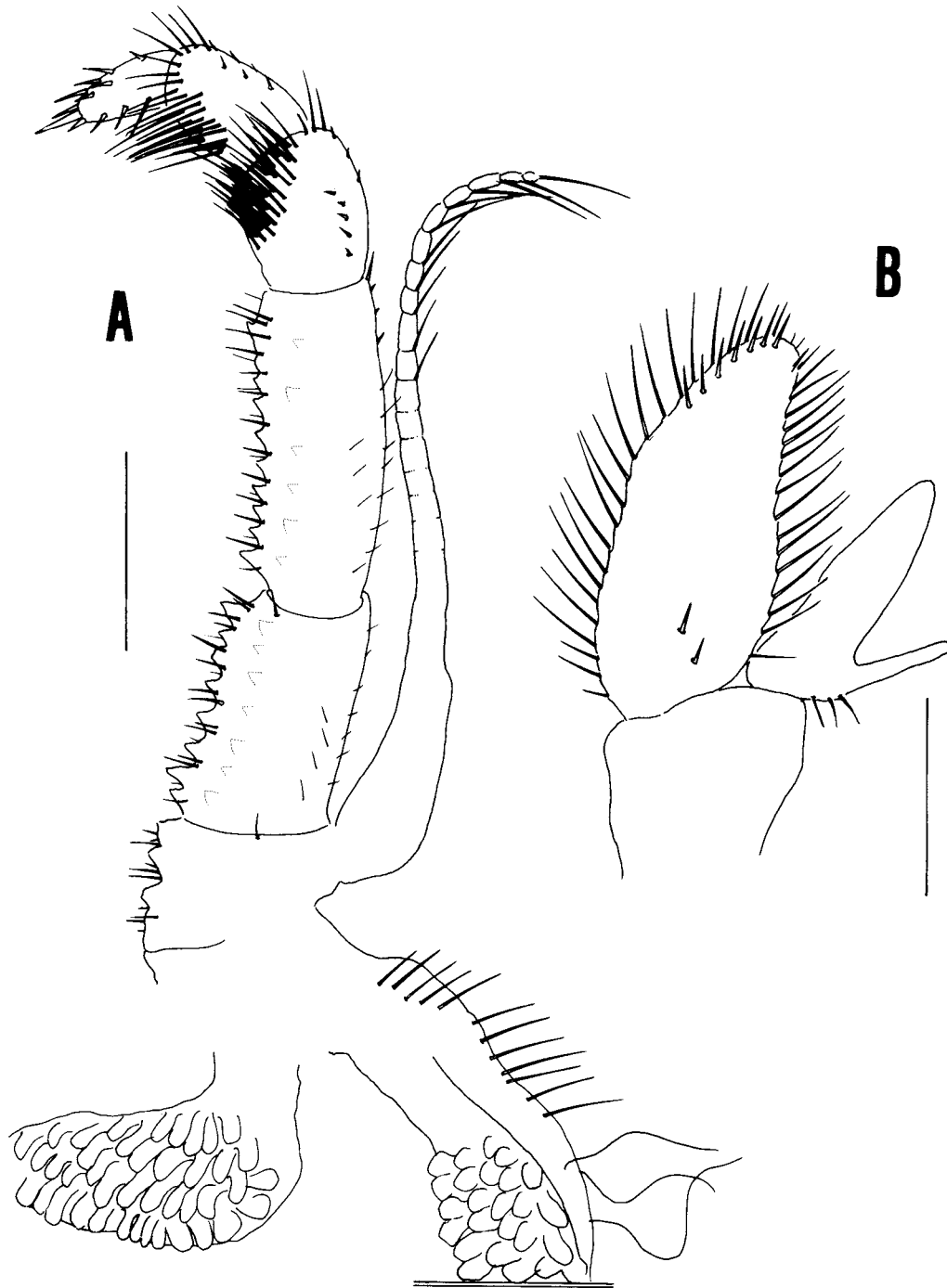


Fig. 9. *Palinurus mauritanicus* Gruvel, 1911. Early juvenile. A, third maxilliped; B, pleopod 4. Scale bars of A = 1 mm, scale bar of B = 0.5 mm.

groups (McWilliam, 1995). However, the puerulus stage of the species which the adult presents a flagellated exopod in the second maxilliped or in the second and third maxillipeds, do not present this character (Table 2). Therefore, the puerulus of *Palinurus mauritanicus*, *P. elephas*, *Projasus bahamondei* and *P. parkeri*, present adult characters in the third maxillipeds that are not present in *Panulirus* species (or in *Jasus edwardsii*) until the juvenile stage is attained.

Also, the thoracic sternum of the puerulus of several *Panulirus* species presents a large posteriorly directed posterolateral spine on the last somite (Deshmukh, 1966; Johnson, 1971; Crosnier, 1971; McWilliam, 1995). However, in *P. mauritanicus* this spine is not directed posteriorly, but is ventrally directed, and additionally, somites 2-4 also bear a spine.

The first juvenile stage is the first fully benthic phase of spiny lobsters and already bears all the distinguishing

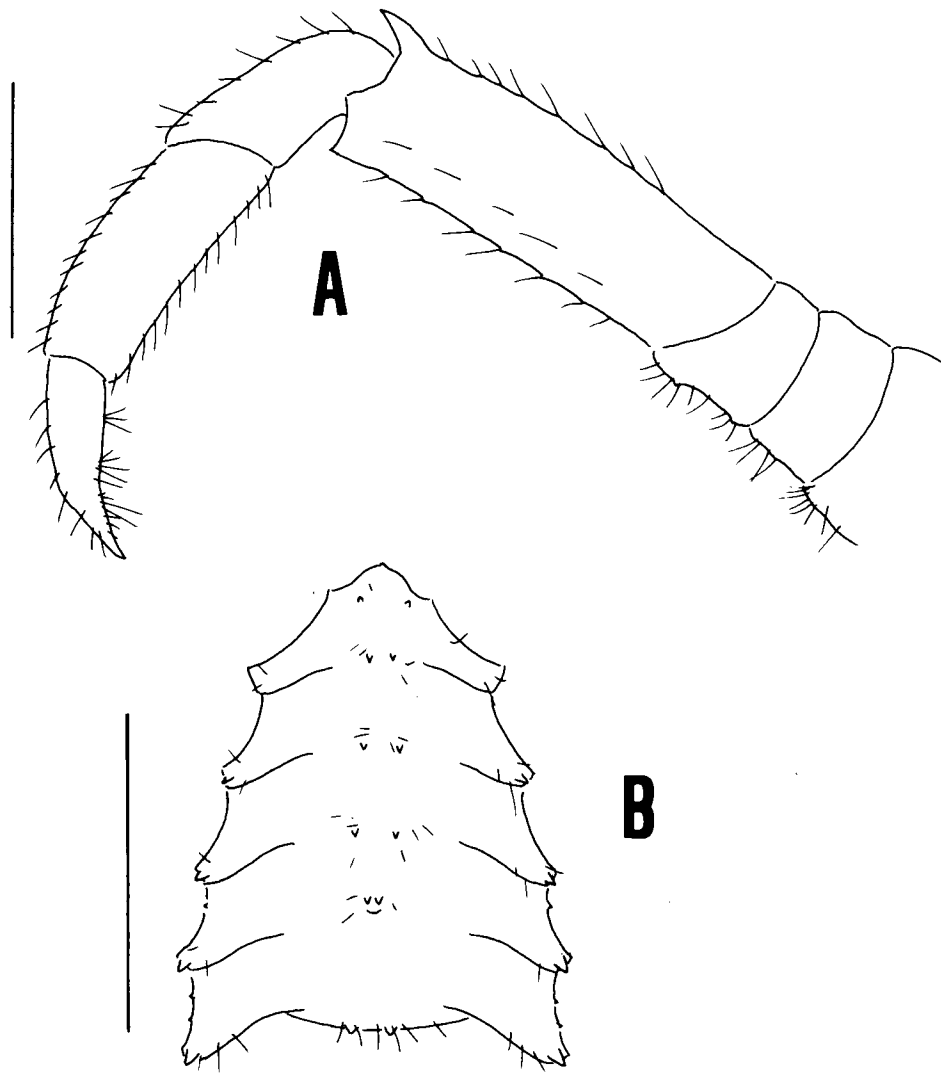


Fig. 10. *Palinurus mauritanicus* Gruvel, 1911. Early juvenile. A, pereopod 1; B, thoracic sternum. Scale bar = 2 mm.

morphological features of the adults (Crosnier, 1971; Cook and McDonald, 1981; Booth and Phillips, 1994; Báez and Ruiz, 2000). Since the size of the early juvenile (specimen C) is practically the same as that of the two pueruli (specimens A and B), this individual would most probably correspond to the first juvenile instar. Overall, the morphology of the early juvenile stage of *P. mauritanicus* described herein is very similar to that of the adults, except for the pigmentation pattern, which might be related to optimization of juvenile survival. Orton and Ford (1933) described the colouration of the early juvenile of *P. elephas* (see also Díaz et al., 2001; González-Pérez, 1995), which, as in *P. mauritanicus*, shows a markedly different pattern to that shown by the adult phase. In *P. mauritanicus*, the arrangement of the spines on the dorsal surface of the carapace is similar in juveniles and adults and can be used to identify the species at all sizes. However, the relative size of the pereopods is slightly different from that of the adult; thus, the shortest pereopod is the first in both juvenile and adult, but is followed in length by the fifth in

the early juvenile and clearly by the second in the adult (Zariquiey-Cenarro, 1935). The developmental changes in the fine structure of the mouthparts from the puerulus to the early juvenile of *P. mauritanicus* are similar to those described for other palinurid lobster species (Nishida et al., 1990; Wolfe and Felgenhauer, 1991; Báez and Ruiz, 2000): increase in setation (in number and complexity), occurrence of rows of teeth (crista dentata) on the third maxilliped, mandible calcification and development of the incisor and molar processes, and elongation of endites of the maxilla.

The early juvenile D captured in the present study, similar in morphology but slightly larger than individual C, may be a later stage, but it could also be a first juvenile stage, since variability in size has been shown in other first juvenile stages (Crosnier, 1971). In fact, García-Raso (1996) reported one juvenile individual of *P. mauritanicus*, 8.5 mm in carapace length, from a depth of 390–400 m, captured by trawling off Mediterranean Morocco, which may well correspond to a first benthic stage of the species.

Table 2. Comparative morphology of the second and third maxillipeds of several palinurid species: *Palinurus mauritanicus* (present study); *P. elephas* (Santucci, 1926); *Panulirus argus* (Wolfe and Felgenhauer, 1991); *P. longipes* (Michel, 1971; Inoue et al., 2002); *P. marginatus* (Rosewater and Kadolsky, 1981); *P. inflatus* (Gordon, 1953); *P. gracilis* (Báez, 1983); *P. guttatus* (Lyons and Hunt, 1997); *P. regius* (Crosnier, 1971); *P. polyphagus* (Deshmukh, 1966); *P. penicillatus* (Michel, 1971); *P. versicolor* Michel (1971); *P. homarus* (Gordon, 1953; Michel, 1971); *Jasus edwardsii* (Nishida et al., 1990); *Palinurellus weinecki* (Michel, 1971); *Projasus parkeri* (Webber and Booth, 1988); *P. bahamondei* (Báez and Ruiz, 2000). +, present; –, absent; +/-, reduced; mxpd, maxilliped; nd, no data.

	Adult exopod 2nd mxpd flagellated	Adult exopod 3rd mxpd flagellated	Puerulus exopod 2nd mxpd flagellated	Puerulus exopod 3rd mxpd flagellated	Puerulus 3rd mxpd Crista dentata
<i>Palinurus mauritanicus</i>	+	+	+	+	+
<i>Palinurus elephas</i>	+	+	+	+	+
<i>Panulirus argus</i>	+	+	–	–	–
<i>Panulirus longipes</i>	+	+	–	–	–
<i>Panulirus marginatus</i>	+	+	–	–	–
<i>Panulirus inflatus</i>	+	–	+	–	nd
<i>Panulirus gracilis</i>	+	–	–	–	–
<i>Panulirus guttatus</i>	+	–	–	–	–
<i>Panulirus regius</i>	+	–	–	–	–
<i>Panulirus polyphagus</i>	+	–	–	–	–
<i>Panulirus penicillatus</i>	+	–	–	–	nd
<i>Panulirus versicolor</i>	+/-	–	–	–	–
<i>Panulirus homarus</i>	–	–	–	–	–
<i>Jasus edwardsii</i>	+	+	+	–	–
<i>Palinurellus weinecki</i>	+	+	+	–	–
<i>Projasus parkeri</i>	+	+	+	+	+
<i>Projasus bahamondei</i>	+	+	+	+	+

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